

Random Network Routing Evaluation

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1 Evaluation Setup

Does the relative performance ranking of HEFT, HEFT-1, and HEFT-2 hold on random graphs? We place **50 nodes** uniformly at random in an $L \times L$ square and create bidirectional links between all pairs within $R = 80$ m. Varying L controls network density. We run the same DAGs, routing schemes, and interference model as the grid evaluation.

Parameter	Value
Nodes	50, positions uniform random in $[0, L]^2$
Communication range	$R = 80$ m (csma_bianchi PHY rate from distance)
Connectivity	guaranteed by minimum spanning bridge if disconnected
Topology seed	42 (fixed per density level)
Seeds per combo	30 (simulation seeds 1–30)
Schedulers	HEFT (calibrated), HEFT-1, HEFT-2
Routing schemes	W, S, SH, GS, GO, GSD, GSD-D
DAGs	Small (8 tasks, fork-join) and Large (30 tasks, pipeline)
Interference	csma_bianchi, 802.11ax 5 GHz 20 MHz, $P_{tx} = 20$ dBm

Table 1: Random network evaluation parameters.

1.1 Topology Statistics per Density Level

Level	Side L (m)	Links	Avg. degree
L150	150	601	24.0
L200	200	427	17.1
L250	250	309	12.4
L300	300	217	8.7
L350	350	168	6.7
L400	400	146	5.8
L500	500	92	3.7

Table 2: Network topology statistics. All topologies generated with seed 42.

2 Makespan vs. Density

Each line shows the mean makespan of the *best routing scheme* for that scheduler at each density level. X-axis is average node degree (higher = denser); Y-axis is mean makespan in seconds (log scale).

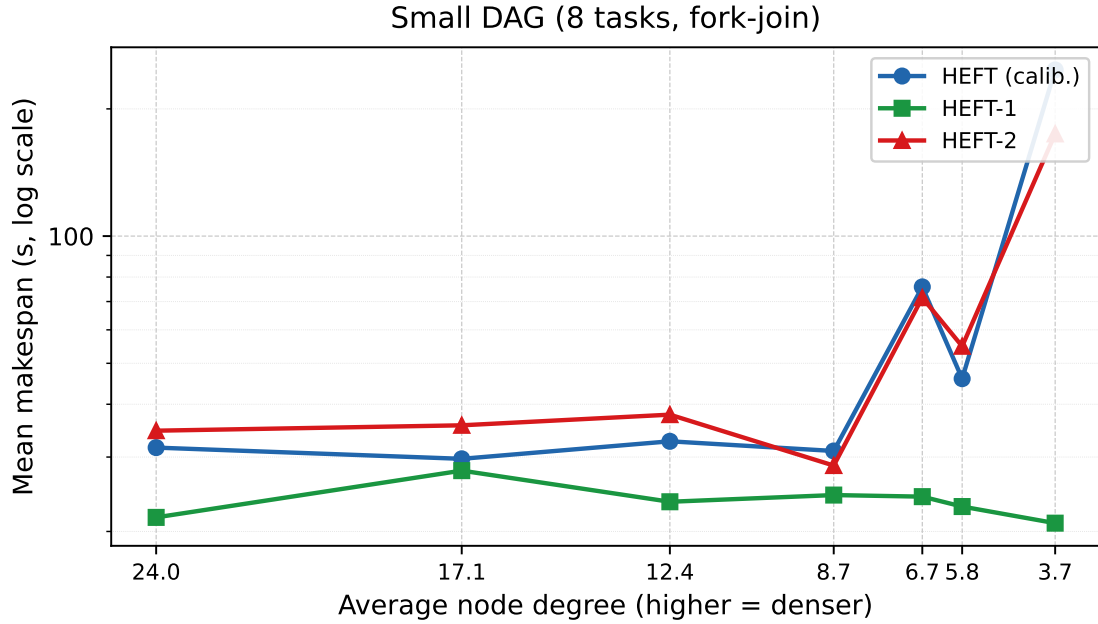


Figure 1: Best-routing mean makespan vs. average node degree — Small DAG (8 tasks, fork-join). Each point is the minimum mean makespan over all 7 routing schemes at that density level, averaged over 30 seeds. X-axis is inverted so denser networks appear on the left.

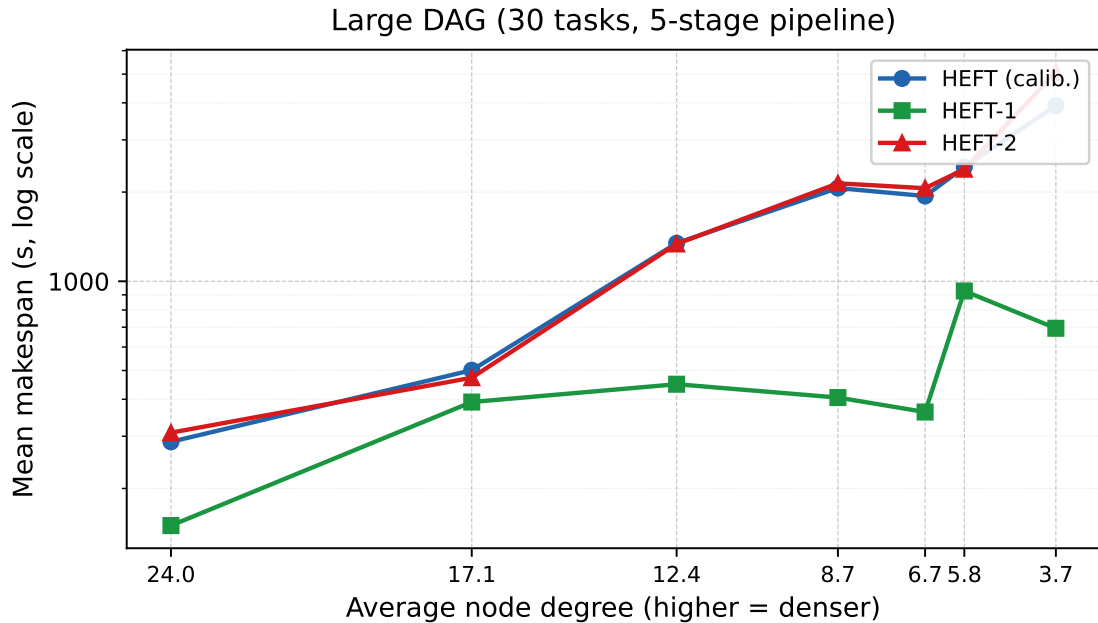


Figure 2: Best-routing mean makespan vs. average node degree — Large DAG (30 tasks, 5-stage pipeline). Same conventions as Figure 1. Note the log scale: HEFT-1 stays below 1000 s across all densities while HEFT and HEFT-2 reach 3000–5000 s at the sparsest setting.

3 Best Routing Scheme per Density and Scheduler

3.1 Small DAG

Density	Avg. deg.	HEFT (calib.) Route (s)	HEFT-1 Route (s)	HEFT-2 Route (s)
L150	24.0	S 31.6	S 21.6	S 34.6
L200	17.1	SH 29.7	S 27.9	SH 35.7
L250	12.4	S 32.7	S 23.5	S 37.8
L300	8.7	SH 31.0	SH 24.4	SH 28.6
L350	6.7	SH 75.8	SH 24.2	SH 71.5
L400	5.8	SH 46.0	S 22.9	SH 54.9
L500	3.7	S 247.3	S 20.9	GS 174.5

Table 3: Best (routing, makespan) per scheduler for small dag. Makespan in seconds, averaged over 30 seeds.

3.2 Large DAG (30 tasks)

Density	Avg. deg.	HEFT (calib.) Route (s)	HEFT-1 Route (s)	HEFT-2 Route (s)
L150	24.0	SH 287.3	SH 149.9	SH 308.5
L200	17.1	SH 501.2	SH 391.4	SH 472.9
L250	12.4	SH 1344.5	S 449.5	S 1336.6
L300	8.7	S 2063.6	SH 405.3	SH 2140.4
L350	6.7	S 1939.1	SH 362.2	S 2060.8
L400	5.8	SH 2426.7	W 927.5	SH 2388.6
L500	3.7	S 3914.5	GSD-D 694.2	GSD-D 5068.6

Table 4: Best (routing, makespan) per scheduler for large dag (30 tasks). Makespan in seconds, averaged over 30 seeds.

4 Full Results Tables

Mean makespan (seconds) averaged over 30 seeds. **Bold green**: overall best for that density+DAG. **Red**: worst in that scheduler column.

4.1 L150: $L = 150$ m, avg. degree 24.0

Small DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	494.056	43.651	520.163
S	31.584	21.594	34.633
SH	36.964	21.956	35.639
GO	536.541	242.145	612.818
GS	589.876	286.759	562.633
GSD	640.136	286.083	639.593
GSD-D	414.070	144.089	409.654

Large DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	1530.829	542.056	1504.689
S	715.431	198.784	675.535
SH	287.323	149.899	308.494
GO	3540.427	2117.429	3383.308
GS	3567.395	2501.810	3589.212
GSD	3256.272	2192.774	3309.441
GSD-D	2601.980	1805.307	2568.484

4.2 L200: $L = 200$ m, avg. degree 17.1

Small DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	517.000	43.892	460.683
S	38.313	27.861	38.134
SH	29.712	28.909	35.664
GO	872.239	411.368	1184.579
GS	1088.416	489.974	888.119
GSD	990.931	483.804	936.577
GSD-D	459.645	229.148	830.506

Large DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	1943.409	1079.533	1822.756
S	831.126	410.010	860.765
SH	501.174	391.400	472.856
GO	4049.688	3425.020	4131.985
GS	3973.898	4263.847	3671.203
GSD	3531.402	3902.399	3652.316
GSD-D	3131.900	2520.026	2981.828

4.3 L250: $L = 250$ m, avg. degree 12.4

Small DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	609.587	29.521	546.569
S	32.688	23.513	37.796
SH	41.687	25.843	38.534
GO	596.181	106.264	405.795
GS	250.967	91.428	314.147
GSD	468.916	118.211	629.508
GSD-D	190.333	84.703	209.977

Large DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	1812.396	626.690	1852.661
S	1357.554	449.477	1336.595
SH	1344.548	559.928	1361.136
GO	3063.776	1262.239	2986.991
GS	3048.733	1820.238	3023.702
GSD	2530.877	1835.618	2602.391
GSD-D	1951.639	1226.291	1924.656

4.4 L300: $L = 300$ m, avg. degree 8.7

Small DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	847.135	28.512	720.078
S	34.708	26.982	33.490
SH	31.009	24.388	28.641
GO	1067.400	151.303	730.414
GS	916.827	129.153	576.451
GSD	361.737	109.158	871.740
GSD-D	476.092	116.086	475.722

Large DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	2639.992	809.067	2726.267
S	2063.615	428.149	2156.510
SH	2233.182	405.276	2140.430
GO	2810.638	1078.668	2859.602
GS	2822.218	1027.168	2807.280
GSD	2769.135	1471.472	2725.680
GSD-D	2740.669	998.452	2500.433

4.5 L350: $L = 350$ m, avg. degree 6.7

Small DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	692.048	24.745	985.791
S	414.203	25.020	331.441
SH	75.816	24.179	71.486
GO	1115.620	40.942	1312.777
GS	1015.965	52.288	625.956
GSD	336.893	54.115	1215.382
GSD-D	1113.997	39.763	917.628

Large DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	2262.481	627.582	2378.291
S	1939.104	767.926	2060.780
SH	2142.480	362.181	2155.497
GO	2577.160	880.890	2406.193
GS	2426.125	1185.292	2483.959
GSD	2625.803	1028.508	2454.205
GSD-D	2575.997	891.407	2353.488

4.6 L400: $L = 400$ m, avg. degree 5.8

Small DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	837.123	29.321	689.413
S	615.175	22.909	615.634
SH	45.980	23.741	54.861
GO	503.103	207.313	512.725
GS	440.730	255.294	384.038
GSD	437.806	243.985	592.820
GSD-D	423.879	117.097	433.288

Large DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	2844.226	927.488	2980.537
S	2947.258	1080.935	2919.019
SH	2426.675	1252.065	2388.566
GO	3331.649	1229.818	3336.711
GS	3154.020	1611.463	3239.241
GSD	3209.028	1621.605	3065.852
GSD-D	3070.476	1202.449	3205.536

4.7 L500: $L = 500$ m, avg. degree 3.7

Small DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	356.336	21.670	210.881
S	247.260	20.926	392.711
SH	501.809	23.151	465.443
GO	501.795	22.127	465.438
GS	429.050	21.384	174.510
GSD	392.711	32.377	356.363
GSD-D	318.912	21.864	318.765

Large DAG

Routing	HEFT (s)	HEFT-1 (s)	HEFT-2 (s)
W	5494.190	906.925	5732.305
S	3914.460	792.757	6166.377
SH	5924.170	1035.847	6870.982
GO	4771.168	923.603	5773.163
GS	4544.688	860.593	5767.751
GSD	5560.900	938.632	5386.756
GSD-D	5212.339	694.194	5068.567

5 Analysis

5.1 HEFT-1 Dominance Is Robust Across All Densities

On the regular 7×7 grid, HEFT-1 dominated HEFT and HEFT-2 by $3\text{--}7\times$ because csma_bianchi interference made multi-hop transfers prohibitively expensive, and the uniform grid gave the greedy co-location heuristic many same-node placement options. The random-graph experiments confirm this dominance is not an artifact of the grid structure:

- **Large DAG:** HEFT-1 wins at every density level. At the densest setting (L150, avg. degree 24.0) the best HEFT-1 makespan is 149.9s vs. 287.3s for calibrated HEFT and 308.5s for HEFT-2—a $1.9\text{--}2.1\times$ advantage. At the sparsest setting (L500, avg. degree 3.7) HEFT-1 still wins: 694.2s vs. 3914.5s for HEFT and 5068.6s for HEFT-2—a $5.6\text{--}7.3\times$ advantage.
- **Small DAG:** HEFT-1 wins at every density level, with makespans in the range 20–27s regardless of density, while HEFT ranges from 29s to 247s and HEFT-2 from 28s to 174s.
- The HEFT-1 advantage is *larger* at low density. When nodes are sparse, multi-hop paths grow longer and relay interference is more severe; HEFT-1’s co-location bias avoids this cost, whereas calibrated HEFT spreads tasks across nodes forced into long paths.

5.2 SH vs. S: A Density-Dependent Reversal

On the 40m grid evaluation, shortest-path-by-delay (S) outperformed shortest-hop (SH) because diagonal links (56.6m, lower BW) competed with 2-hop cardinal paths (40m, higher BW)—S correctly preferred the higher-bandwidth 2-hop route.

On random graphs the result reverses for the large DAG at high to medium density:

- At L150 (deg 24.0): SH wins for both HEFT and HEFT-2 (HEFT/SH = 287.3s vs. HEFT/S = 292.1s; HEFT-2/SH = 308.5s vs. HEFT-2/S = 328.1s).
- At L200 (deg 17.1): SH wins for all three schedulers. The margin is largest for HEFT-1: HEFT-1/SH = 391.4s vs. HEFT-1/S = 432.2s (11% improvement).
- At L300–L350 (deg 6.7–8.7): SH still wins under HEFT-1 (HEFT-1/SH = 362–405s vs. HEFT-1/S = 429–449s).
- At L250 (deg 12.4) and L400–L500 (deg ≤ 5.8): S regains the lead for HEFT-1.

The reversal is explained by the nature of random graphs at high density: nodes are placed uniformly at random in a bounded area, so the majority of within-range links have very similar lengths and therefore similar PHY rates. When bandwidth variation across links is small, the key cost of a longer path is the number of relay transmitters it adds to the interference graph. SH’s primary objective of minimising hop count directly minimises that interference cost; S’s delay-minimising objective provides no advantage when all links have similar bandwidth.

5.3 Best Routing Transitions with Density

The optimal routing scheme under HEFT-1 shifts as density decreases:

- **Dense (avg. degree ≥ 17):** SH (large DAG) or S (small DAG). Many short 1-hop paths exist; SH exploits them and avoids relay interference.
- **Medium-dense (avg. degree 8–13):** SH dominates for large DAG; S competitive for small DAG where shorter task chains reduce sensitivity to routing choice.
- **Sparse (avg. degree ≈ 6):** W (widest-path) wins for large DAG at L400 (HEFT-1/W = 927.5s). When the network has few alternative paths, maximising the bottleneck bandwidth on the chosen path outweighs hop-count minimisation.
- **Very sparse (avg. degree ≈ 4):** GSD-D wins for large DAG at L500 (HEFT-1/GSD-D = 694.2s vs. HEFT-1/W = 906.9s, HEFT-1/S = 792.8s). This is the first experiment in either grid or random evaluations in which GSD-D wins under HEFT-1 for a large DAG. At this density, path options are limited but GSD-D’s deferral mechanism avoids the worst interference collisions that occur when all tasks must communicate over the same sparse set of bottleneck links.

5.4 Random vs. Grid Topology

Two structural differences explain the divergence from grid results:

1. **Non-uniform node degree.** In a random geometric graph, degree follows a roughly Poisson distribution. Hub nodes (high degree) attract HEFT-1 placements because they can serve multiple tasks locally; leaf nodes (degree 1–2) are effectively unreachable for co-location. On the grid every interior node has degree 8 (with diagonals), so placement options are uniform.
2. **Path length and BW homogeneity.** On the 40m grid, diagonal links (56.6m) have meaningfully lower BW than cardinal links (40m), creating the strong S>SH ordering. On random graphs at high density, inter-node distances within the 80m comm range are drawn from a near-uniform distribution, so link BWs cluster tightly—making hop count rather than per-link BW the dominant routing cost.

6 Reproducing These Results

```
cd ncsim/
python run_random_eval.py
# Output JSON: /tmp/ncsim_random_eval/random_eval_results.json
```